

Does R-Lens Produce More Contextually Coherent Early-Layer Readouts? A Blinded Two-Model Evaluation

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CAMBRIA capstone — replication of the R-Lens coherence claim

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Abstract

R-Lens is a variant of J-Lens that fits its layer-to-final-layer transport with LRP stop-gradients. Its authors report, qualitatively and without a released scorer, that R-Lens readouts at early layers are more coherent than J-Lens readouts. We test that claim directly on two 27B models with a blinded, pre-specified and frozen rating panel of 200 three-arm cells, scored by two independently validated LLM autoraters with a third-judge adjudication rule. R-Lens is rated more contextually coherent than J-Lens on both models: +0.655, 95% CI [0.480, 0.835] on Qwen3.5-27B and +0.900, [0.625, 1.170] on Gemma-3-27B-it, on a 0–4 scale. The direction holds under every one of four scoring rules and under restriction to cells where both lenses echo the prompt equally, which retains roughly two thirds of the pooled effect. Three findings qualify this. R-Lens also scores significantly higher on prompt echo, and echo coincides with about a third of the measured gap. Comparisons against the logit-lens baseline are *not* stable across judges—on Qwen the two primary judges disagree in sign about whether J-Lens beats the logit lens at all. And all three lenses score low in absolute terms: the best arm averages 2.07–2.39 out of 4. We replicate the ordering, not a claim that early-layer R-Lens readouts are coherent.

1 What this document does and does not claim

This is the coherence replication only. It asks one question: when a human-like rater is shown three unlabelled early-layer readouts of the same prompt, one from the logit lens, one from J-Lens and one from R-Lens, does the R-Lens arm receive a higher contextual-coherence score?

It makes **no** claim about concept specificity, causal onset, ablation, necessity, sufficiency, answer smuggling, or whether any lens recovers a representation the model actually uses. Separate experiments in this project address some of those; by pre-specification their results are not pooled with, used to interpret, or reported alongside the coherence estimand, and they do not appear here.

Terminology. The protocol was written, frozen and hashed before any rating was collected, but it was not deposited with a third-party registry. We therefore say *pre-specified and frozen* throughout, never *preregistered*.

2 Artifacts and provenance

Both lenses are the authors’ released artifacts, not refits, taken at a pinned commit. A preflight check refuses to run if either lens fails provenance:

- J-Lens carries {"estimator": "standard"}.

- R-Lens carries {"estimator": "relp"} with the LN rule, identity rule and half rule at $\beta = 0.5$.
- **Both were fit with n_prompts: 25**—the released light recipe, not the $n = 1000$ recipe described in the source post. Any difference between our effect size and theirs is confounded with this.
- The two Jacobian sets are verified to actually differ, so a null could not be produced by silently loading the same lens twice.

Preflight additionally verifies hook indexing (that recorded block outputs satisfy `activations[ $\ell$ ]` equals `hidden_states[ $\ell+1$ ]`), target layer identity, and transport orientation. A failure at any of these is fail-closed: the run aborts rather than proceeding with a warning.

3 Method

3.1 Panel

Twenty prompts—four from each of the five official evaluation sets (association, multihop, multilingual, poetry, typo)—were selected by a hash of a fixed salt, the set name and a stable item identifier, so selection is reproducible and independent of any result. For the two sets carrying a target (multihop, multilingual) only items the model itself answers correctly were eligible, matching the source protocol’s correctness filter. Prompts are shared between models: both models are read on the same 20 items.

Each prompt is read at five pre-specified relative depths $z = \ell/\ell^* \in \{0, 0.1, 0.2, 0.3, 0.4\}$, giving $20 \times 5 \times 2 = 200$ cells, each holding three arms. Readout position follows the source protocol: the final prompt token, except poetry, read at the newline ending line one.

3.2 Blinding

Arm order is permuted per (*cell*, *rater*) pair, not once per cell. A single fixed order would make position bias identical across judges and therefore invisible to any agreement statistic. The unblinding key was written outside the repository and outside the published result tree; the panel and the sample carry content hashes; and every outgoing judge payload was reconstructed after the fact and scanned for structural leakage of lens identity.

3.3 Rubric

Three dimensions per arm: **contextual coherence** (0–4, primary), **lexical integrity** (0–2) and **prompt echo** (0–2). Contextual coherence asks whether the readout tokens form a plausible continuation or elaboration of the prompt’s content. Prompt echo measures how much of the readout is copied from the prompt — it is scored so that copying can be detected, not rewarded.

3.4 Judges

Two primary autoraters, `openai/gpt-5` and `deepseek/deepseek-chat-v3.1`, each independently validated on a 78-cell battery of control categories before being admitted. Validation gates include position-bias rate (tested with an exact binomial, so that a threshold finer than the achievable resolution at that sample size returns UNDERPOWERED rather than a false pass), a corruption-detection control, a duplicate-arm control, and a response-completeness gate that fails a judge

returning empty content on more than 2% of cells. One candidate judge was rejected under exactly this last gate.

When the two primaries disagree, a third judge (`meta-llama/llama-3.1-70b-instruct`) rates the cell and the score becomes the median of three; otherwise it is the mean of two.

The adjudicator has not yet been put through the same 78-cell validation battery. It touched 63% of cells, so this is a live gap. Section 5.2 shows what the result looks like without it.

3.5 Statistics

The estimand is the equal-weight paired difference: mean over evaluation sets, then prompts within set, then depths within prompt, so that no set or prompt is over-weighted by having more cells. Intervals are percentile 95% from a 10,000 replicate stratified paired *prompt-cluster* bootstrap. p -values are paired prompt-cluster sign-flip permutation tests — never a bootstrap sign proportion. Depth and set breakdowns are Holm-corrected within model.

4 Results

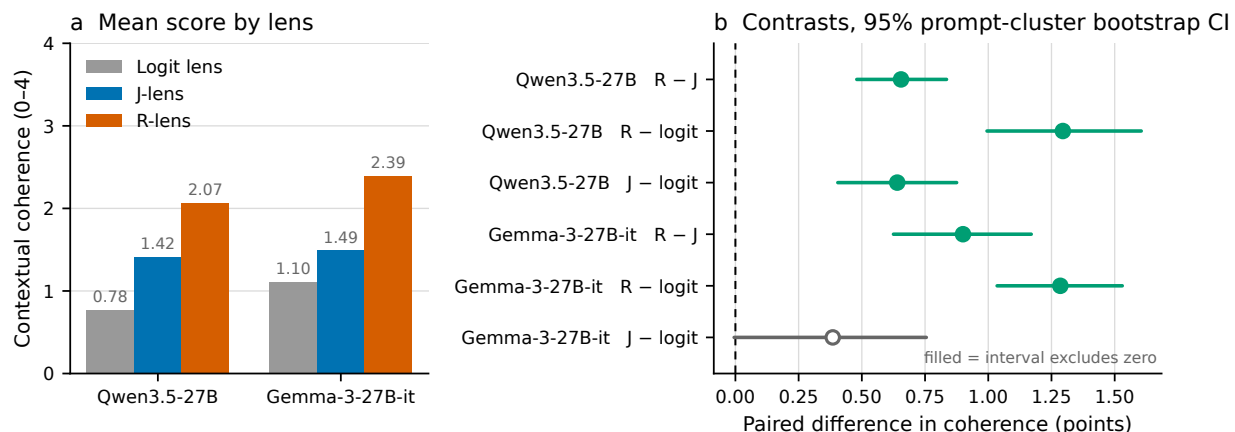


Figure 1: Primary result. (a) Mean contextual coherence by lens on the full 0–4 rubric scale. (b) Paired contrasts with 95% prompt-cluster bootstrap intervals; a filled marker denotes an interval excluding zero. The axis in (a) is deliberately not rescaled: the ordering is clear, but every lens sits in the lower half of the scale.

4.1 Primary endpoint

R-Lens is rated more contextually coherent than J-Lens on both models (Table 1). Both intervals exclude zero in the positive direction, so the ordering claim replicates on both models.

Two features of this table matter as much as the headline.

On Gemma, J-Lens is not distinguishable from the logit lens (+0.385, interval touching zero, $p = 0.126$). The advantage on that model is carried by the R-Lens variant, not by the Jacobian transport as such.

Absolute scores are low. Mean contextual coherence is 2.07 (Qwen) and 2.39 (Gemma) for R-Lens against 1.42 and 1.49 for J-Lens (Figure 1a). On this rubric the mid-scale band is “partially relevant, not a coherent continuation”. R-Lens readouts at $z \leq 0.4$ are better than the alternatives and still not good.

Model	Contrast	Δ	95% CI	p	win	tie	loss
Qwen3.5-27B	R – J	0.655	[0.480, 0.835]	$< 10^{-4}$	.61	.27	.12
Qwen3.5-27B	R – logit	1.295	[0.995, 1.605]	$< 10^{-4}$	.80	.09	.11
Qwen3.5-27B	J – logit	0.640	[0.405, 0.875]	0.0003	.61	.22	.17
Gemma-3-27B-it	R – J	0.900	[0.625, 1.170]	0.0004	.63	.24	.13
Gemma-3-27B-it	R – logit	1.285	[1.035, 1.530]	$< 10^{-4}$	.75	.08	.17
Gemma-3-27B-it	J – logit	0.385	[-0.005, 0.755]	0.126	.53	.11	.36

Table 1: Primary endpoint and the two supporting contrasts. 100 cells and 20 prompts per model. Positive Δ favours the first arm.

4.2 Depth

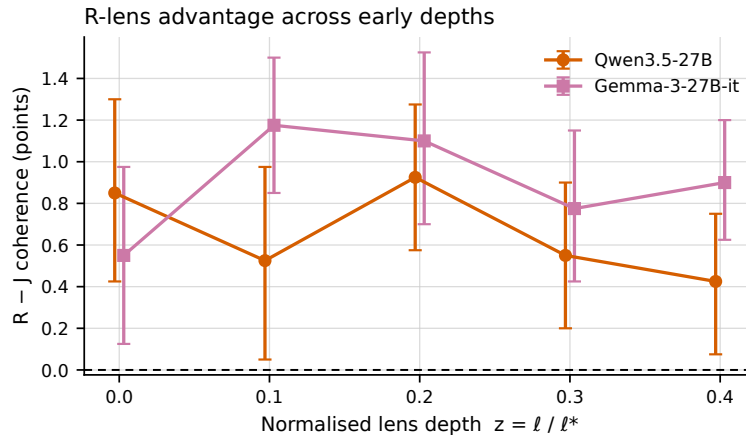


Figure 2: R – J by normalised depth, with 95% intervals. The two models do not agree on where in depth the advantage is largest.

The depth profiles disagree between models. On Qwen the effect is largest at the embedding layer ($z = 0$, +0.850) and decays monotonically to +0.425 by $z = 0.4$. On Gemma the effect at $z = 0$ is the *weakest* of the five depths (+0.550) and the only one that does not survive Holm correction ($p_{\text{Holm}} = 0.092$); it peaks at $z = 0.1$ (+1.175).

A reading on which R-Lens specifically repairs the earliest layers is supported by Qwen and contradicted by Gemma. What both models support is the weaker statement that R-Lens is ahead somewhere across $z \leq 0.4$.

4.3 Secondary dimensions

Model	Dimension (R – J)	Δ	95% CI
Qwen3.5-27B	Lexical integrity	0.125	[0.060, 0.190]
Qwen3.5-27B	Prompt echo	0.225	[0.135, 0.300]
Gemma-3-27B-it	Lexical integrity	0.165	[0.035, 0.305]
Gemma-3-27B-it	Prompt echo	0.305	[0.210, 0.395]

Table 2: Secondary dimensions, both on 0–2 scales.

R-Lens produces more well-formed tokens (lexical integrity) *and* copies more of the prompt (prompt echo). Both differences exclude zero on both models. Section 5.3 asks whether the second explains the first.

We do not compare the magnitude of the echo difference to the magnitude of the coherence difference. They are measured on different scales (0–2 and 0–4) and such a comparison would be meaningless.

4.4 By evaluation set

Per-set differences are all positive on both models and range from +0.450 (Qwen, typo) to +1.900 (Gemma, poetry). **No per-set difference survives Holm correction** ($p_{\text{Holm}} \geq 0.636$ throughout). With four prompts per set this analysis is descriptive only and is reported for pattern, not for inference.

5 Robustness

5.1 Integrity audit

A fail-closed audit re-derived the frozen artifacts from scratch: it recomputed the panel hash, replayed the per-(cell, rater) arm permutation and compared it to the key, reconstructed all 400 outgoing judge payloads and scanned them for lens-identity leakage, and re-parsed every stored judge response.

Twenty conditions were checked. Nineteen passed, including zero unparseable responses across 526 stored ratings, zero arm-mapping mismatches, and an exact panel-hash reproduction. One failed: the score-combination rule derived the combined winner by *voting* on the judges’ winners while *averaging* their scores, which let the two disagree on 18 of 200 cells. The rule was corrected to derive the winner from the combined scores, and the frozen raw responses were re-combined without any new API call. **Re-running the full analysis on the corrected scores changed no reported number**—all 18 cells were ties under both rules.

5.2 Judge dependence

The primary estimate used a third judge on 63% of cells, so the headline is a function of the scoring rule as well as of the lenses. Recomputing under four frozen rules (GPT-5 alone; DeepSeek alone; mean of the two primaries with no adjudication; the frozen adjudicated rule) gives a positive R – J on every rule and on both models, with every individual-judge interval excluding zero. The pooled estimate ranges from +0.725 (DeepSeek alone) to +0.975 (GPT-5 alone), with the adjudicated rule at +0.778. **Adjudication made the estimate smaller, not larger** (mean-of-two pooled: +0.850), so the third judge was a conservative influence rather than the source of the effect.

The logit-lens contrasts are not stable. That verdict is scoped to R – J, and the scoping is load-bearing. Checking every contrast against the same two single-judge intervals, **five of nine are not stable across the two primary judges**. On Qwen, GPT-5 rates J-Lens *significantly worse* than the logit lens (−0.680, [−1.06, −0.33]) while DeepSeek rates it *significantly better* (+0.770, [0.57, 0.97])—same cells, disjoint intervals, opposite signs. Every unstable contrast involves the logit lens; every R – J contrast is stable.

The natural reading is a floor effect: the judges agree about the relative ordering of two Jacobian transports and anchor the bottom of a 0–4 scale differently when asked to score near-noise. Any

R-lens advantage is present under every scoring rule

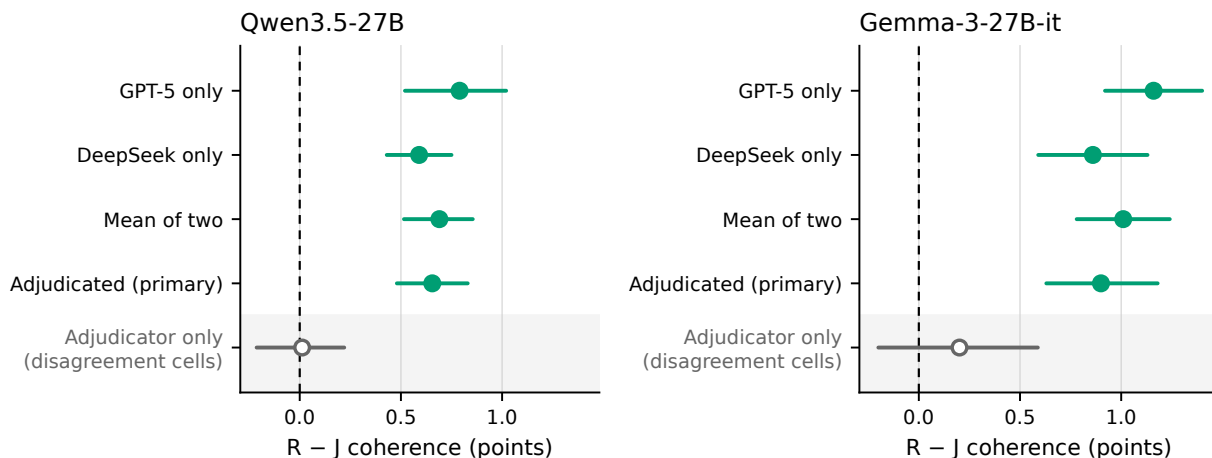


Figure 3: R - J under four scoring rules. The shaded band is a diagnostic computed only on cells where the two primary judges disagreed; it is conditioned on disagreement and is not a fifth estimate of the same quantity.

comparison against the logit lens in this report should be read with both judges’ values in view and must not be quoted as a single number.

Agreement. Quadratic-weighted $\kappa = 0.514$ on 600 paired scores, exact agreement 39.3%, mean absolute difference 0.91 points, and a 63% adjudication rate. This is moderate agreement at best. With validated judges it is evidence about the material rather than the instrument—the readouts being rated are genuinely ambiguous—but it is not a strong instrument, and **it is not fully validated while the adjudicator’s own battery is outstanding.**

5.3 Prompt echo

R-Lens copies more of the prompt than J-Lens, and a rater who rewards prompt-shaped text would score it higher for that reason alone. Restricting to cells where both lenses received the *same* echo score:

Model	All cells		Echo equal retained		Echo both zero retained	
Qwen3.5-27B	0.655	0.467 [0.25, 0.70] ($n=74$)	71%	0.453 [0.21, 0.70] ($n=71$)	69%	
Gemma-3-27B-it	0.900	0.841 [0.58, 1.13] ($n=60$)	93%	0.788 [0.50, 1.11] ($n=51$)	88%	
Pooled	0.778	0.523 [0.35, 0.70] ($n=134$)	67%	0.518 [0.33, 0.71] ($n=122$)	67%	

Table 3: R - J contextual coherence under echo restriction, frozen adjudicated scoring. “Retained” is the restricted estimate as a percentage of the all-cells estimate.

The effect survives in all 24 subset estimates—every scoring variant, both models, both restriction rules—with intervals excluding zero throughout. **But roughly a third of the pooled gap coincides with an echo difference**, and the retention is markedly higher on Gemma (93%) than on Qwen (71%).

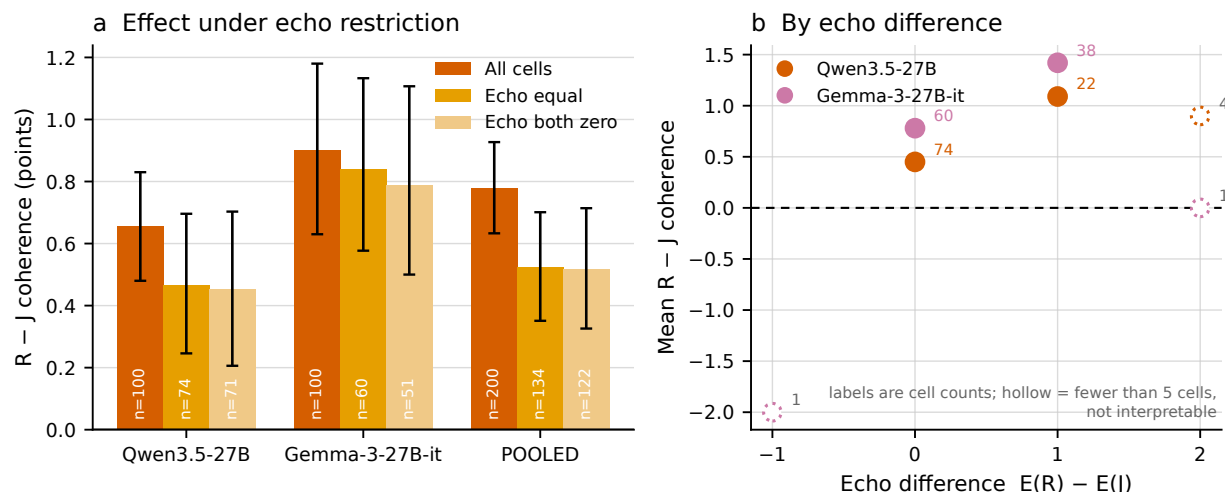


Figure 4: Echo sensitivity. (a) $R - J$ restricted by echo agreement, retained cell counts printed inside each bar. (b) Mean $R - J$ by echo difference; labels are cell counts and hollow markers mark strata too small to interpret.

This is not a causal adjustment. Echo is measured from the same readouts as coherence, so restricting on it conditions on a post-treatment variable. The regression of the coherence difference on the echo difference is descriptive; its intercept is the fitted difference at equal echo, not an echo-adjusted effect. **The two judges also disagree about the mechanism:** GPT-5’s slope is ≈ 1.0 – 1.1 with intervals excluding zero on both models, while DeepSeek’s is 0.29 $[-0.16, 0.80]$ and 0.35 $[-0.03, 0.63]$, both covering zero. GPT-5 sees echo and coherence tightly coupled; DeepSeek sees almost no coupling. Both still rate R above J .

What would settle it. A rubric that scores coherence *excluding* copied spans, validated against controls in which a pure prompt-copy arm must not win, would test this directly rather than by restriction. That instrument is specified but not yet run.

5.4 Judge behaviour

R-Lens wins 61–63% of $R - J$ cells and loses 12–13%, with the remaining quarter tied. The effect is a broad shift, not a handful of extreme cells.

6 The earlier no-prompt panel

An earlier 50-cell panel on Qwen returned a null for $R - J$ ($+0.080$, 95% CI $[-0.16, 0.34]$, $p = 0.596$). It is reported here for completeness and is **not comparable** to the present result. The two panels differ simultaneously in prompt visibility (that panel did not show raters the prompt), size (50 vs 200 cells), rubric scale, judge count, sampling rule and adjudication, and one of its raters was contaminated. No single one of those differences can be identified as the reason the results differ, and we do not claim to know which mattered. It is a preliminary attempt that was superseded, not a contradicting measurement.

That panel did produce one durable negative result, obtained against 150 human judgements: of three cheap token-form proxies for coherence, only the “trash-rate” proxy correlated in the expected

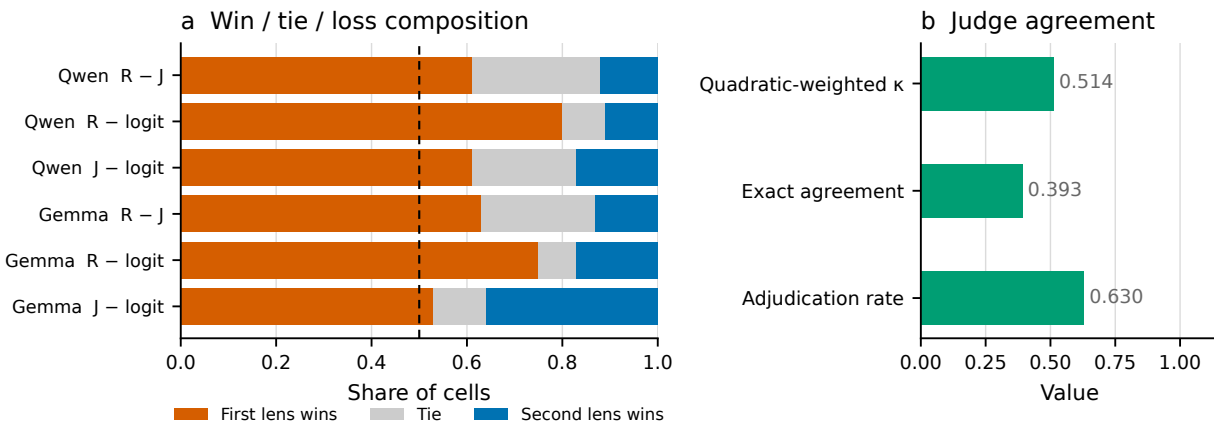


Figure 5: What the judges did. (a) Win/tie/loss composition per contrast. (b) Agreement statistics. A mean difference of 0.78 points is compatible with very different underlying distributions; the composition shows which.

direction ($\rho = -0.640$). Zero-corpus-frequency rate ($\rho = +0.330$) and in-prompt rate ($\rho = -0.245$) were both *inverted*, retracting two earlier readings of ours. Token-form statistics are not a substitute for rating the readouts.

7 Limitations

1. **The adjudicator is not validated.** It touched 63% of cells. The result holds without it (mean-of-two: +0.850 pooled; both single judges significant on both models), but until its battery is run the primary scoring rule rests on an unvalidated component.
2. **Released lenses use the 25-prompt light recipe**, not the 1000-prompt recipe of the source post. Effect sizes are not comparable to the authors’.
3. **Twenty prompts per model.** Adequate for the pooled estimand, which is why intervals are prompt-clustered; not adequate for the per-set breakdown, which is descriptive only.
4. **Autoraters, not humans.** No human panel was run. Judges were validated against controls, but LLM raters may share biases with each other and with the models being read.
5. **Echo is not causally controlled**, only restricted on.
6. **Logit-lens comparisons are judge-dependent** and should not be quoted as single numbers.
7. **Depth-localisation does not replicate across models.**
8. **Ordering, not quality.** Nothing here shows early-layer R-Lens readouts are coherent in absolute terms; they average roughly 2 of 4.

8 What would change the conclusion

- The adjudicator failing its validation battery would not overturn the direction, but would demote the adjudicated estimate and promote the mean-of-two.
- A non-echo rubric—coherence scored excluding copied spans—returning a null would move the reading from “R-Lens reads out more coherently” to “R-Lens copies the prompt more, and raters reward that”.

- A human panel disagreeing with the autoraters at scale would invalidate the instrument, not merely widen the intervals.
- Refitting both lenses at $n = 1000$ and finding the gap disappears would indicate the effect is a property of the light recipe.

9 Reproducibility

Every number in this document is produced by a command in the repository operating on frozen artifacts; none is transcribed by hand into a plotting script or a table. The figures are drawn by `rlens figures` directly from the analysis outputs, and skip themselves by name if an input is missing rather than falling back to defaults.

Stage	Command
Preflight (fail-closed)	<code>rlens preflight</code>
Prompt selection	<code>rlens eligibility</code>
Panel freeze	<code>rlens freeze-panel, rlens panel-v2</code>
Judge validation	<code>rlens judge-validate</code>
Rating	<code>rlens autorate</code>
Primary analysis	<code>rlens analyse-v2</code>
Integrity audit	<code>rlens audit-v2</code>
Judge and echo sensitivity	<code>rlens robustness</code>
Figures	<code>rlens figures</code>

Artifact hashes, the protocol salt, seeds and the audit verdicts are recorded in `final_reproducibility_manifest.json`. The unblinding key is deliberately absent from the repository and from this document’s source tree.